

02LQD Specification and Simulation of Digital Systems
2021/09/03

1. Combinational circuit design (2 points)

Design a combinational converter from BCD to excess-3 code. Input X on 4 bits is a BCD digit. Output Z on 4 bits is the corresponding excess-3 code. The excess-3 code for a decimal digit is the result of adding 3 to the decimal digit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

An FSM has a 1-bit input X and a 1-bit output Z. It serves as a sequential converter from BCD to excess-3 code. The BCD input arrives least significant bit (LSB) first. After 4 inputs the circuit resets to the initial state ready for another group of 4 inputs. The excess 3 code is output serially at the same time LSB first as well..

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Design the Mealy FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (16 points)

A system counts the number of events on a single-bit input B and always outputs that number unsigned on a 6-bit output C, which is initially 0. An event is a change from 0 to 1 or from 1 to 0. Assume that system count rolls over when the maximum value of C is reached. Reset is asynchronous.

1. Draw the state diagram of a HLSM that captures the above behavior (**6 points**)
2. write the corresponding VHDL code (**6 points**).
3. Design a FSM-D (**4 points**):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Monday September 6 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.